JHNSRE

The JHU Near-Space Radiation Experiment

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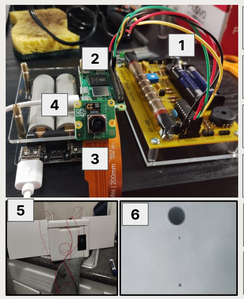
With thanks to Fiore True Value Hardware, and the students of Bishop Guilfoyle Academy.

We accidentally did plasma physics in the stratosphere.

35,000 m above sea level.

Radiation permeates the atmosphere at all altitudes: as cosmic rays collide with atmospheric nuclei, cascades of secondary particles are produced. These showers peak in intensity at approximately **15–25 km** in a phenomenon known as the **Pfotzer maximum**. In this regime, the air density requirement to drive cascade effects **balances** with a minimum of overlying atmosphere.

JHNSRE is an inexpensive, high-altitude balloon apparatus built to characterize this maximum, and to photograph Earth from the middle stratospheric region:



1. MightyOhm Geiger PCB
SBM-20 tube, 1.5V AAA, 400V HV+ lead

2. Raspberry Pi Zero 2W
OpenBSD 7.8, 32GB SD, gzipped logs

3. Camera + Ribbon Cable
Raspberry Pi Camera Module 3W

4. Molicel P28A 18650 ×2
5V series, Li-Ion, UPS HAT, to Raspberry Pi

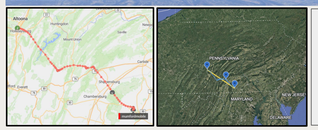
5. Enclosure, Satellite Antenna
2x GPS trackers, total mass 699.9 ±0.1 (g)

6. Kaymont 1600 Balloon
4400L He, 2500g chute, ascent rate 5.6 m/s

JHNSRE was launched from Altoona, PA on 2026/04/24 following FAA clearance (#ZOB04-622) with the help of ~100 K-5 Bishop Guilfoyle Academy students. **The launch vehicle performed as expected**, traveling through a novel corridor over the Appalachian mountains. The trajectory taken closely followed pre-flight models developed by Cambridge University, with payload recovery near Gettysburg, PA.

Result: Peak altitude attained = 35,500 ± 150 (m)

Expected value = 35,000 ± 500 (m) (exact) // t = 0.96

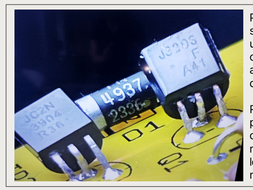


120 ± 1 (km)
Distance travelled

3:01
Flight duration

Our science payload, however, experienced multifactorial failure. The fragile ribbon cable first sustained mechanical shock in an aborted launch attempt - rendering high-altitude photography impossible.

Following this, as in-flight pressure decreased with altitude, we strongly believe the Geiger counter high-voltage bias circuit underwent Paschen breakdown - masking the Pfotzer curve.



Post-flight inspection of the flyback switching transistor (Q1, TO-92, upper right) reveals physical damage: carbonized epoxy and bronzed leads are consistent with frequent discharges from the collector.

Repeated breakdown near 400V produced EMI coupled to the detector, resulting in anomalous measurements. We thus report a lower bound for the Pfotzer maximum, prior to instrument failure.

Result: Pfotzer maximum ≥ 15,500 ± 300 (m)

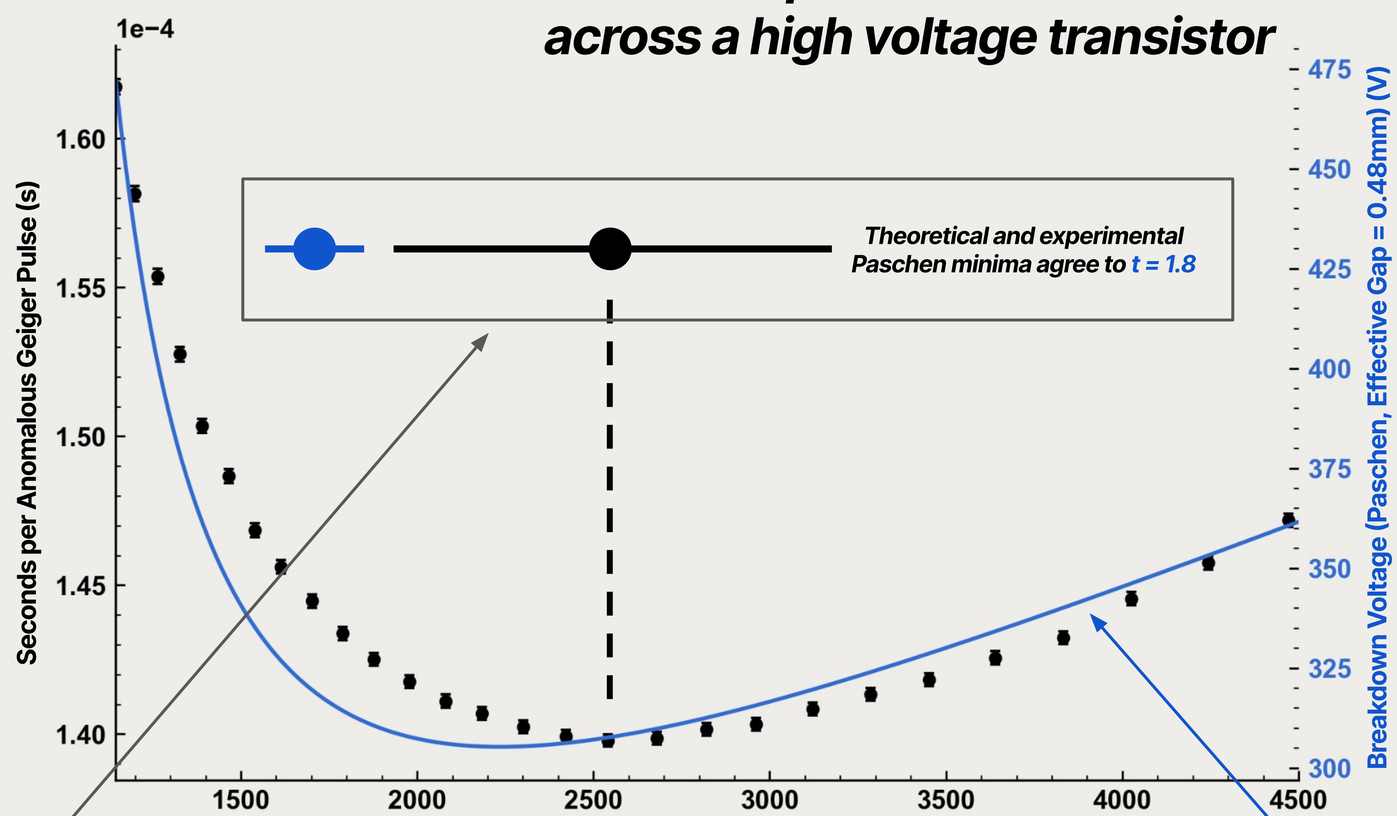
Expected value = 20,000 ± 5,000 (m) // plausible bound

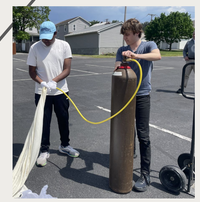
Beyond this bound, we **re-interpret high count rates as EMI events originating at Q1.** This produces the two well-characterized Paschen breakdown curves to right, symmetric on ascent and descent. These demonstrate reproducible Paschen minima, in agreement with theory:

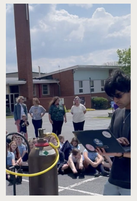
Result: Paschen Minimum = 2.5 ± 0.8 (Pa • m)

Parallel plate assumption = 1.12 ± 0.19 (Pa • m) // t = 1.8

EMI events as proxies for ionization across a high voltage transistor



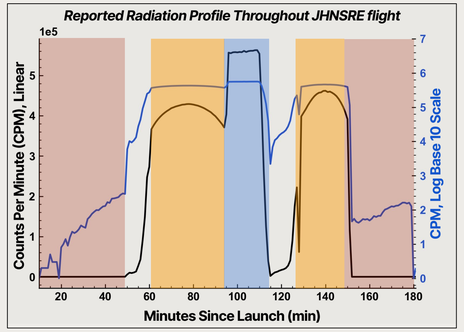




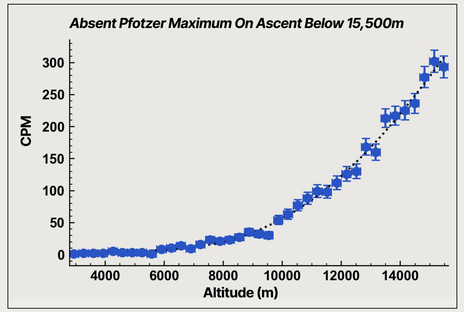
"We cannot guarantee that this goes perfectly. As our instructor says, we are experimentalists..."

~ Jay, hours before launch, to a group of fifth graders

Our flight data sees enormous variation in recorded count frequencies. We extract three dominant regimes: the **Pfotzer Regime (RED)**, the **Paschen Regime (ORANGE)**, and the **Short Regime (BLUE)**.



Beyond ~50 minutes, CPM measurements exceed the SBM-20's maximum physical count rate (~2 μs dead time), and therefore cannot originate from genuine ionization in the tube. This is a **dominant source of uncertainty** which we can't mask by plotting a partial Pfotzer curve:



Connection to the Physics: The Paschen Regime

Paschen's Law gives the breakdown voltage across two parallel plates:

$$V_B = \frac{Bpd}{\ln(Apd) - \ln\left[\ln\left(1 + \frac{1}{\gamma_{se}}\right)\right]}$$

Differentiation followed by rearrangement identifies an optimal value, which is **prominently** visible in the experimental curves above.

$$pd = \frac{e}{A}\ln\left(1 + \frac{1}{\gamma_{se}}\right)$$

Here **A** and **B** are well-documented properties of the medium (i.e. air), while **γ** reflects the quantity of secondary electrons produced per positive ion. We therefore identify this theoretical optimum for air:

γ
Values from 0.001 to 0.2 are reported. Assume **0.01 ± 0.008**

Known A, B →

Paschen Minimum
1.12 ± 0.19 (Pa • m)

...and take **γ** as uncertain.

$$\delta\gamma_{re} = \frac{d(pd)}{d\gamma_{re}}\gamma_{re} = \frac{e}{A\gamma_{re}(\gamma_{re}+1)}$$

To verify this in experiment, **we crucially assume that the time-averaged count rate *monotonically increases* in breakdown rate at the discharge site:**

Lower breakdown voltage | **More EMI bursts per second** | **Higher CPM**

The inverse CPM (or seconds-per-count) curve thus inherits the shape of the breakdown voltage - though it is distorted by several unknowns, the minimum remains intact and can be extracted:

Altitude
Known via GPS and Cambridge University model to **±150 meters**

Collector-Emitter Gap
Measured as **1 ± 0.3 mm**

↓

Pressure
Tabular conversion through a European Space Agency dataset

→

Paschen Minimum
2.5 ± 0.8 (Pa • m); t = 1.8

The experimental value is high, with some residual - consistent with **(1.)** non-uniform electric fields at the leads, and **(2.)** and progressive surface tracking shortening the inter-gap distance on the package.

As further confirmation of our theory, we run a free-parameter fit of the Paschen curve over our data. **This fit is shown to left in blue**, beautifully capturing our detector's behavior, and converges to an effective parallel plate gap distance of **0.5mm.**